Command

Command · Undoable editor keystrokes · Build with Waqar

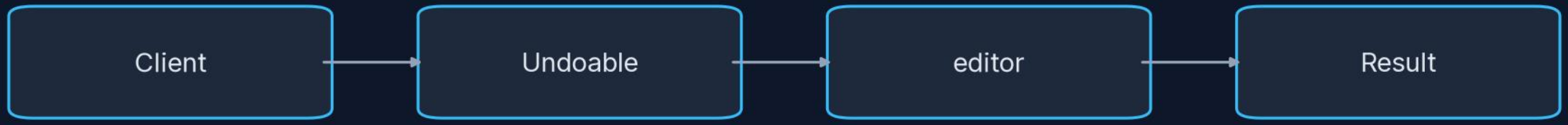
Today's Hook

- Problem: Undoable editor keystrokes
- Pattern: Command — earned from the pain, not trivia.
- Runnable main on JDK 25 with captured stdout.
- Naive → solution → failure paths → trade-offs.

The Problem

- Undoable editor keystrokes
- Naive design shows the cost clearly.
- Pattern isolates the force behind a clean boundary.
- Tests cover happy path and injected failures.

Structure



Command: Undoable editor keystrokes

Real-World Scenario

- Running example: Undoable editor keystrokes
- Same names in naive code, refactor, and Demo output.
- No shared StreamForge framing unless it fits.
- Product language first; pattern vocabulary second.

Naive Code

```java
package com.buildwithwaqar.patterns.gof.behavioral.command;

import com.buildwithwaqar.patterns.gof.behavioral.command.*;
/** Problem: text editor needs undoable typing. */
public final class UseCaseDemo {
  public static void main(String[] a) {
    System.out.println("=== Command problem: undoable editor ===");
    var ed = new Editor(); var bus = new CommandBus();
    bus.run(new WriteCommand(ed, "Hello "));
    bus.run(new WriteCommand(ed, "Waqar"));
    System.out.println("text=" + ed.text());
    bus.undo();
    System.out.println("undo=" + ed.text());
  }
}
```

Why Naive Hurts

- Call sites couple to the wrong shape or leak on partial failure.
- Parameter drilling / duplicated envelopes / missing rollback.
- Hard to unit-test without the real dependency.
- New variants require shotgun edits.

Solution

```java
package com.buildwithwaqar.patterns.gof.behavioral.command;

import java.util.ArrayDeque;
import java.util.Deque;

interface Command { void execute(); void undo(); }

final class Editor {
    private final StringBuilder text = new StringBuilder();
    void append(String s) { text.append(s); }
    void deleteLast(int n) { text.delete(Math.max(0, text.length() - n), text.length()); }
    String text() { return text.toString(); }
}

record WriteCommand(Editor editor, String chunk) implements Command {
    public void execute() { editor.append(chunk); }
    public void undo() { editor.deleteLast(chunk.length()); }
}

/** Invoker with undo stack. */
public final class CommandBus {
    private final Deque<Command> history = new ArrayDeque<>();
```

Solution (2)

```
    public void run(Command command) {
        command.execute();
        history.push(command);
    }

    public void undo() {
        if (!history.isEmpty()) history.pop().undo();
    }
}
```

Runnable main

```java
package com.buildwithwaqar.patterns.gof.behavioral.command;

import com.buildwithwaqar.patterns.gof.behavioral.command.*;
/** Problem: text editor needs undoable typing. */
public final class UseCaseDemo {
  public static void main(String[] a) {
    System.out.println("=== Command problem: undoable editor ===");
    var ed = new Editor(); var bus = new CommandBus();
    bus.run(new WriteCommand(ed, "Hello "));
    bus.run(new WriteCommand(ed, "Waqar"));
    System.out.println("text=" + ed.text());
    bus.undo();
    System.out.println("undo=" + ed.text());
  }
}
```

Walkthrough Output

```
=== Command problem: undoable editor ===
text=Hello Waqar
undo=Hello
```

Step-by-Step Trace

- Each stdout line maps to a stage above.
- Re-run the main class locally on JDK 25.
- Failure-path lines prove compensations / rejects / missing files.
- Pause and predict the next line before revealing.

Modern Java 25 / Zazr

```
// Java 25 / Zazr direction
// records + sealed + switch
// inject Env / collaborators at the edge
// virtual threads for blocking fan-out
```

Trade-offs

- Clarity vs ceremony — refuse five types for two lines.
- Teaching demos omit production metrics/idempotency keys.
- Prefer simpler language features when they erase the force.
- Know the twin pattern often confused with Command.

Related Patterns

- Contrast Command with its neighbor pattern.
- Name the roles from memory in sixty seconds.
- Point to the season map for what comes next.
- Link the GitHub episode folder in the description.

Recap

- Problem → naive → Command → Demo stdout.
- Failure paths are part of the lesson.
- Teach when NOT to use it.
- Practice: break a test, fix it, re-run the main.